

birdclef-2026

10th solution, simple model as always.

Rank	10
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Topic ID	704271
Version	Original

Source: <https://www.kaggle.com/competitions/birdclef-2026/discussion/704271>

Retrieved with Kaggle CLI on 2026-07-03.

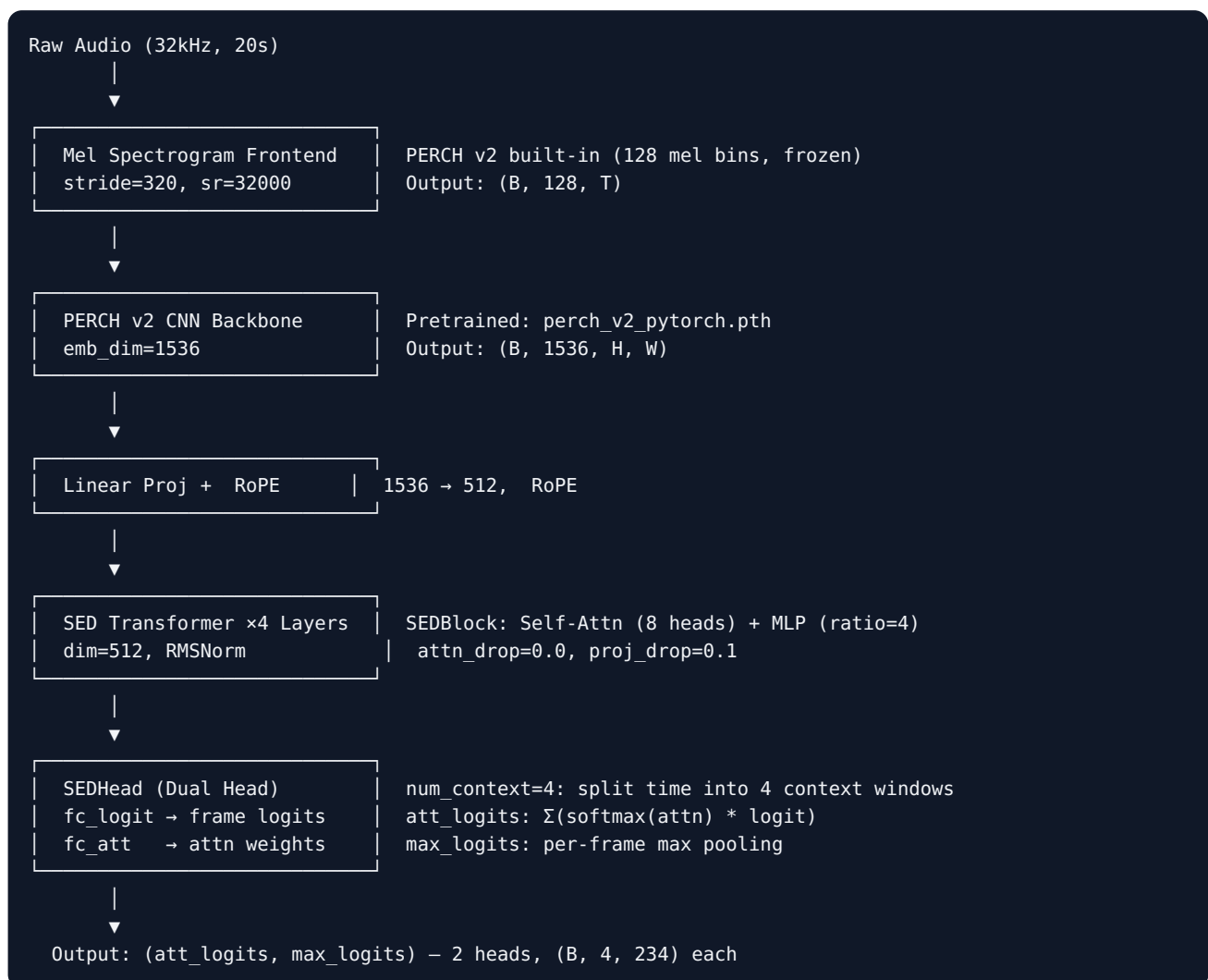
Scheme Summary:

*** The pipeline does not work well with models other than PERCH v2. I have tried several alternative backbones, HGNET as an example. Just like PERCH v2, these models require pre-trained weights. I experimented using BirdSet XC data yet only reached approximately 0.94. ***

1. my best private single model,

<https://www.kaggle.com/code/cooolz/perchv2-sed>

2. Model Architecture



3. Trainable perchv2

The pytorch model is convert from tf version with the help of gpt. And there is a public availabel one produced by [hengck23](#)

4. Train params

Cos scheduel with warmup from 0 to $2.5e-4$ to $1e-6$ with around 10 epochs. Bce loss

5. Sample strategy,

80-500, if a label has morethan 500 files, give a low sample weight. if a label has less than 80 files, give a high sample weight. strong soundscape data x2.5 weight

6. augmantation,

- 0.8 probability: random gain ± 4 dB
- 0.5 probability: random time and frequency masking
- 1.0 probability: mixup

7. Inference & Post-processing

The final inference uses ONNX-exported model via OpenVINO. Post-processing pipeline:

```
60s audio → Mel Spectrogram → Single Model → Prior Fusion
→ Smooth Day/Night → Sigmoid
→ Amphibia Month Prior → File Confidence Scale
→ Per-Taxon Temporal Smoothing → Final Probs
```

7.1. Inference

- 5-fold ONNX models
- Input: 60s audio → 6000-frame mel spec → model → $12 \times 5s$ clip logits
- Output: logits (12, 234)

7.2. post process

7.2.1. Prior Fusion

- Site, hour, site $\times$ hour priors fitted from soundscape strong labels

7.2.2. Smooth Day/Night Correction

Many species follow unique day-night activity rhythms, analogous to higher cold drink sales in summer. Their hour-wise priors are computed from probability distributions of soundscape labels; supporting evidence is shown below.

7.2.3. Amphibia Month Prior

Amphibian activity fluctuates between dry and wet seasons, similar to seasonal beverage consumption trends. Monthly priors are fitted from soundscape probability statistics, with relevant evidence attached below.

This echoes a well-known ML anecdote: a DL expert fitted beverage store data and concluded cold drinks sell more in summer and hot drinks sell better in winter.

7.2.4. File-Level Confidence Scale

7.2.5. Per-Taxon Temporal Smoothing

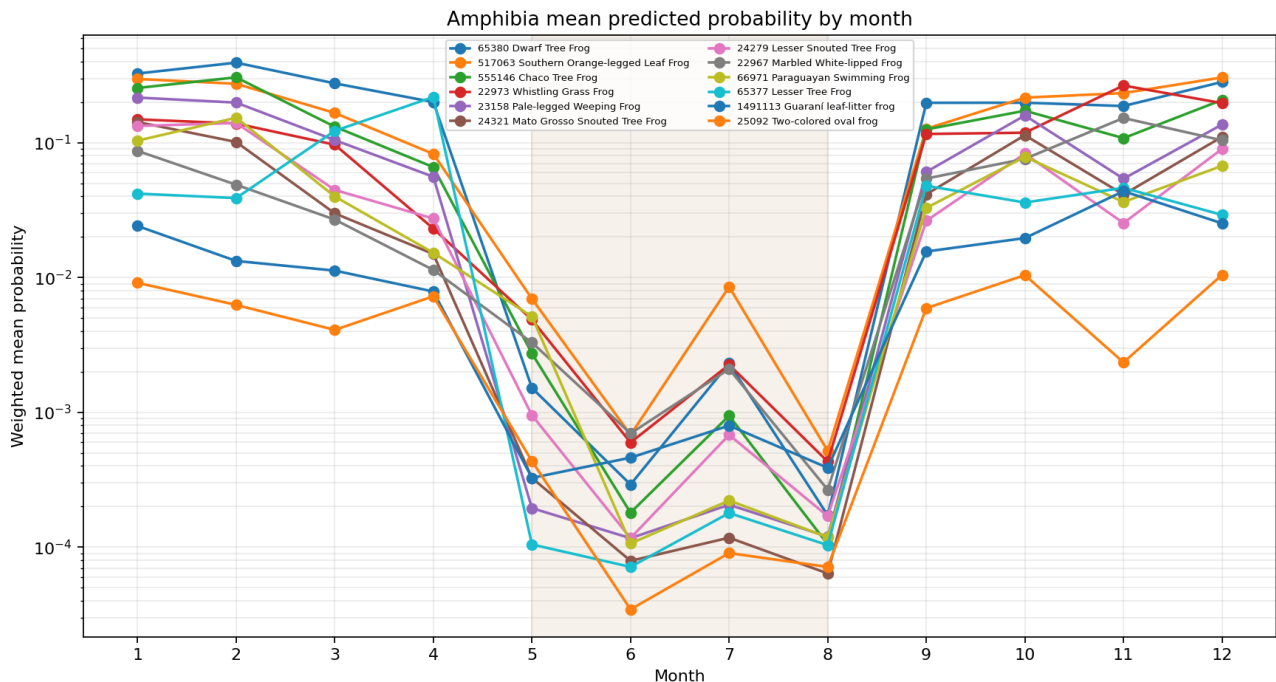
7.3 ensemble

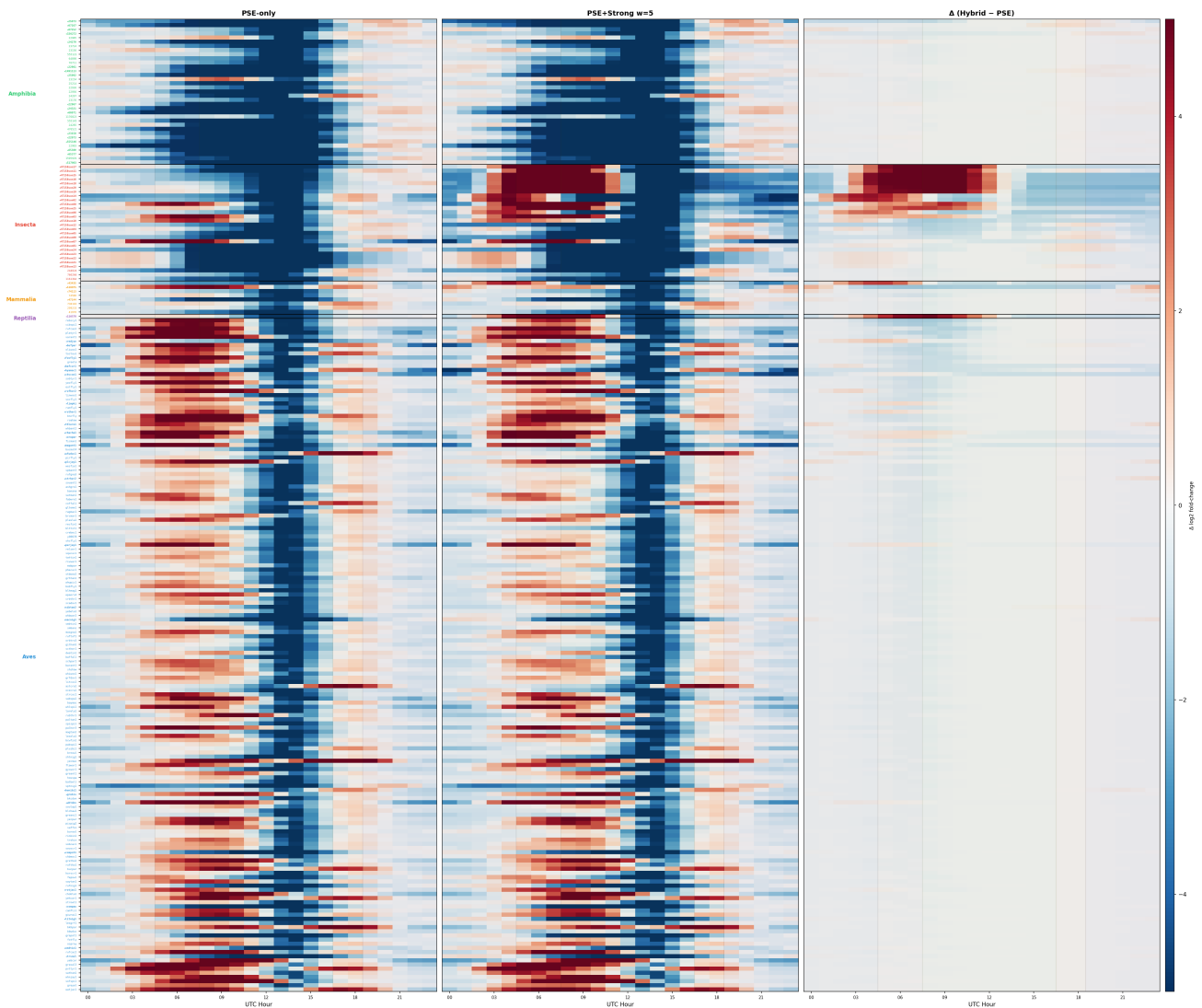
The ensemble lacks diversity owing to exclusive use of the PerchV2 backbone. I substitute the Transformer with Mamba and other sequential models; this modification can maintain similar performance metrics. Actually single model is good enough.

Evidence for the post process,

This time the host gives site, month and hour metainfo, that could be used. I found it late, and got no time to use it within the model.

I got some evidence:





pse label

It remains unclear if pseudo labels benefit the method: a base model trained without pseudo labels scored 0.951 on public LB a month earlier. Regardless, I appended 10~12 pseudo-labeled samples to the training corpus for training.

code share

[birclef26](#) the code is summary by OMO.

Supplemental Q&A

Salman Ahmed • 2026-06-04T14:19:56.653000

[@coolz](#) Would it be possible if you could share your github repo / code for this?

coolz · 2026-06-05T03:53:29.920000

The code was summarized with DeepSeek. Due to AI involvement, the project is disorganized and the overall code flow is hard to trace. Hope this could help.

link, <https://github.com/610265158/birdclef26>